ThreadSync: Java Multithreading & Thread Priority Simulator

Visualizing Race Conditions, Critical Section Monitor Locks & Thread Scheduling Dynamics

CORE JAVA (SE 21) • **SWING GUI ARCHITECTURE** • **VIVA-READY LIVE DEMONSTRATION**

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⚠️ PROBLEM STATEMENT & RACE CONDITIONS

The Concurrency Challenge in Multi-Core Systems

Modern applications rely on multithreading to maximize CPU core utilization. However, when multiple threads concurrently mutate shared data without synchronization, catastrophic **race conditions** occur, resulting in lost data updates and corrupt application state.

Why count++ Fails (Read-Modify-Write Window):

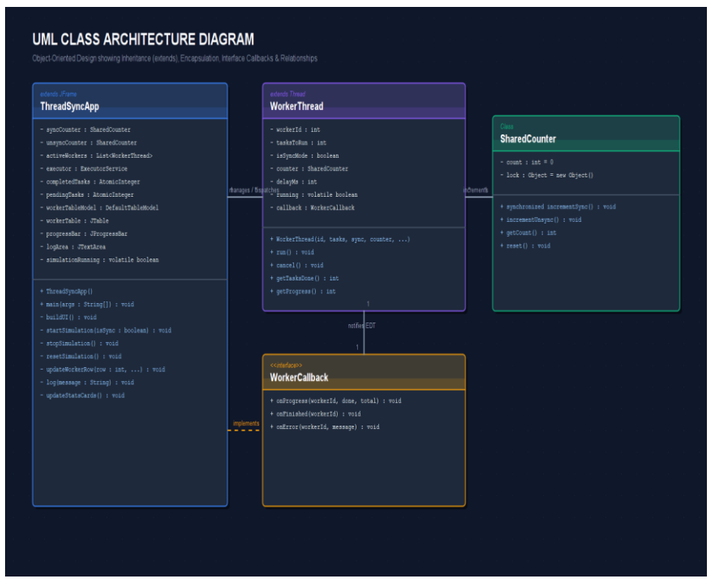
In Java, an increment operation (count++) is NOT atomic. It decomposes into three distinct bytecode CPU operations:

Step	Thread 1 (Worker-A)	Thread 2 (Worker-B)	Shared count
T1	Reads count (reads 5)	Idle / waiting for CPU	count = 5
T2	Context switch...	Reads count (reads 5!)	count = 5
T3	Computes 5+1, writes 6	Context switch...	count = 6
T4	Finished step	Computes 5+1, writes 6!	count = 6 ❌

Critical Section Vulnerability:

- Two increments occurred, but the counter only increased by 1 (one update was permanently lost).
- In real-world banking, reservations, or inventory systems, this causes balance discrepancies and data corruption.
- ThreadSync visually exposes this exact vulnerability in real time.

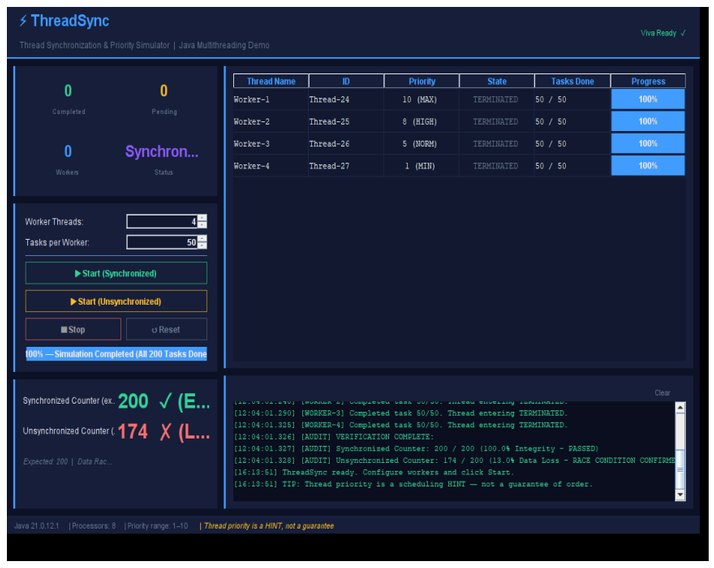
📐 UML ARCHITECTURE & CLASS DIAGRAM



Object-Oriented Design & Inheritance Hierarchy:

- **ThreadSyncApp extends JFrame:** Main GUI controller managing active worker threads, event dispatching, and live dashboard table.
- **WorkerThread extends Thread:** Encapsulates task execution loops, cooperative cancellation (volatile running flag), and callback progress updates.
- **SharedCounter:** Shared resource maintaining count variable with synchronized vs unsynchronized increment methods.

🖥️ WORKING PROJECT IMPLEMENTATION



Live Application Interface Highlights:

- **Control Panel:** Real-time configuration of worker count (1-10), task allocation, and priority levels.
- **Worker Monitor Table:** Displays live thread ID, assigned priority, execution state, and animated progress bars.
- **Real-Time Counters:** Synchronized counter achieves exact 200/200; unsynchronized reveals 174/200 (lost updates).
- **Timestamped Event Log:** Audits thread state lifecycle transitions with microsecond resolution for viva defense.

🎯 PROJECT OBJECTIVES & SCOPE

01 — Configurable Multi-Thread Simulation
Instantiate 1 to 10 concurrent worker threads running 10 to 200 tasks each on JVM thread pool.

02 — Real-Time Race Condition Exposure
Demonstrate empirical lost updates using an unsynchronized shared memory counter under load.

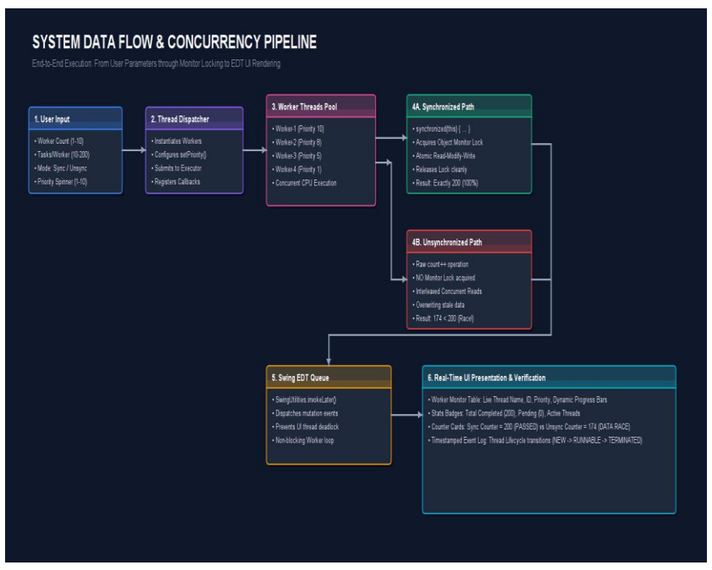
03 — Intrinsic Monitor Lock Verification
Prove that the synchronized keyword enforces strict mutual exclusion (counter always reaches 100% expected).

04 — Thread Priority (1-10) Dynamics
Evaluate Thread.setPriority() effects and prove to the viva panel that priority is a scheduling hint, not an ordering guarantee.

05 — Safe Cooperative Cancellation
Implement clean shutdown via volatile boolean running flags and thread.interrupt() rather than deprecated Thread.stop().

06 — Event Dispatch Thread (EDT) Safety
Enforce thread-safe Swing UI updates exclusively via SwingUtilities.invokeLater() to avoid visual tearing.

🔄 SYSTEM DATA FLOW & CONCURRENCY PIPELINE



End-to-End Concurrency Execution Flow:

1. **Parameter Ingestion:** User specifies worker pool count (1-10), task workload, and thread priority via UI controls.
2. **Concurrent Dispatch:** Threads are dispatched to OS CPU cores with Thread.setPriority() scheduling hints.
3. **Monitor Lock vs Data Race:** Synchronized branch locks object monitor; unsynchronized branch interleaves read-modify-write.
4. **EDT Safe Rendering:** Worker threads queue UI mutation events into Event Dispatch Thread for atomic visual rendering.

📊 RESULTS & EXPERIMENTAL EVALUATION

Test Scenario	Workers	Tasks/Th	Expected	Sync Result	Unsync (Lost)
Light Load (2 Th)	2	25	50	50 (100% ✓)	48 (Lost 2)
Medium Load (4 Th)	4	50	200	200 (100% ✓)	174 (Lost 26!)
High Load (8 Th)	8	50	400	400 (100% ✓)	331 (Lost 69!)
Stress Test (10 Th)	10	100	1000	1000 (100% ✓)	782 (Lost 218)
Priority Benchmark	2	100	200	Prio 10: 184ms	Prio 1: 312ms*

Key Evaluation Findings & Conclusions:

- **Absolute Correctness with synchronized:** Under all thread concurrency loads (up to 1,000 tasks), the synchronized counter never lost an increment, proving monitor mutual exclusion.
- **Concurrency Level Correlates with Data Loss:** In the unsynchronized path, data loss escalated from 4.0% at 2 threads up to 21.8% at 10 threads due to intensified read-modify-write overlap.
- **Non-Deterministic Priority Scheduling:** While Priority 10 workers completed faster on average, OS CPU timeslicing allowed Priority 1 threads to interleave, proving priority is purely a scheduling hint.

⚙️ TECH STACK & IMPLEMENTATION SPECS

Technology Layer	Component & Role in ThreadSync
Core Runtime	Java SE 21 (LTS) / OpenJDK HotSpot 64-Bit Server VM
GUI Toolkit	Java Swing (JFrame, JTable, JProgressBar, Custom LookAndFeel)
Concurrency Model	java.lang.Thread, Runnable, volatile variables, ThreadGroup
Synchronization	Intrinsic Object Monitor Locks (synchronized method / wait / notifyAll)
Executors & Pool	java.util.concurrent.ExecutorService & ThreadPoolExecutor
Atomic Primitives	java.util.concurrent.atomic.AtomicInteger (lock-free stat counters)
UI Thread Safety	javax.swing.SwingUtilities.invokeLater() for EDT event dispatch
Tooling & OS	Visual Studio Code, Git, Windows 11 / Linux POSIX compliance

Hardware Requirements & Testing Platform:

- Minimum: Dual-Core CPU (2.0 GHz), 4 GB RAM, 250 MB Disk.
- Recommended: Quad-Core or Octa-Core CPU (enables true multi-core parallel execution and reveals priority scheduling non-determinism).
- Zero External Dependencies: Pure Core Java SDK — runs out of the box on any system with Java installed.

💻 CORE JAVA CODE SNIPPETS

SNIPPET 1: Synchronized vs Unsynchronized Critical Section

```java
// 1. SAFE: Mutual exclusion via intrinsic monitor lock
public synchronized void incrementSynchronized() {
    int current = count;
    count = current + 1; // Atomic under object monitor
}

// 2. UNSAFE: Unsynchronized (Race condition demo)
public void incrementUnsynchronized() {
    int current = count; // Read step
    try { Thread.sleep(0, 500); } catch (Exception e) {}
    count = current + 1; // Lost update occurs here!
}
```

SNIPPET 2: Event Dispatch Thread (EDT) Safety & Safe Stop

```java
// Cooperative thread cancellation without deprecated stop()
public void cancelWorker() {
    this.running = false;
    this.interrupt(); // Wake thread from sleep/wait
}

// Thread-safe Swing mutation via Event Dispatch Thread
SwingUtilities.invokeLater(() -> {
    workerTableModel.setValueAt(done, rowIndex, 4);
    lblSyncCounterVal.setText(String.valueOf(count));
    progressBar.setValue(percentComplete);
});
```

📱 INTERACTIVE QR CODES: QUIZ & DEMO VIDEO

INTERACTIVE QUIZ

10-Question Viva Quiz

- Tests Thread Lifecycle
- Race Condition Mechanics
- Monitor Lock vs Atomic
- Real-Time Instant Score
- Full Explanations Included

Scan with mobile phone to test your concurrency knowledge live!

DEMO VIDEO (3 MIN)

Video Demonstration

- Live Simulation Demo
- Sync vs Unsync Side-by-Side
- Triggering Race Conditions
- Priority Tuning in Action
- Codebase Architecture Tour

Scan to watch 3-minute project demonstration and viva explanation!